

Properties of Matter

* Elasticity

- (i) Deforming force :- It is an external force when applied to a body it changes shape and size of the object.
- (ii) Restoring force :- It is developed within the body when external force (deforming force) is applied on an object it is also called internal & restoring force which help body (object) to regain original shape and size after removal of deforming force.
- (iii) Elasticity :- It is the property of the solid object by virtue of which it regain its original shape and size after removal of deforming force.

Classification of body according to elastic property

- (a) Perfectly elastic body :- The body which completely regain its shape and size after the removal of deforming force is called perfectly elastic body.

eg - Quartz, phosphor, Bronze.

NOTE - NO body in the universe is perfectly elastic.

- (b) Perfectly Plastic (Inelastic) Body :- The body which (have any tendency) does not regain its original shape and size.

is called after the removal of deforming force is called perfectly plastic body.
eg:- Clay, Paraffin, wax.

NOTE - NO body in the universe is perfectly plastic.

(c) Elastic Body:- Body which regain its original shape and size after removal of deforming force is called elastic body.
eg- Steel, Rubber etc.

(d) Plastic (inelastic) Body:- The body which does not regain its original shape and size after the removal of deforming force is called Plastic Body.
eg- Brass, putty etc.

Date
04/10/24

Calculate Percentage error in the measurement of density of cube and mass of cube

Date
04/10/24

CLASSTIME Pg No.
Date / /

* Stress :- The external deforming force or internal restoring force developed per unit area is called stress.

$$\text{i.e. Stress} = \frac{\text{Deforming or restoring force}}{A}$$

$$= \frac{F}{A}$$

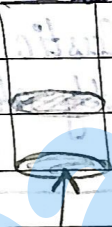
$$\text{SI unit of stress} = \text{Nm}^{-2} = \frac{\text{kg ms}^{-2}}{\text{m}^2} = \text{kgm}^{-1}\text{s}^{-2}$$

$$\text{Dimension of stress} = [M^1 L^{-1} T^{-2}]$$

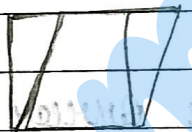
* Types of stress :-

(i) Tensile or longitudinal or Normal $\rightarrow$

(ii) compressive $\rightarrow$



(iii) Tangential $\rightarrow$



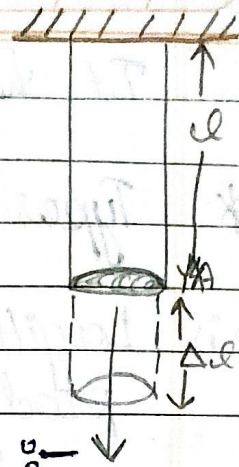
(The stress developed in a body depends upon how the external forces are applied over it)

There are three types of stress :-

(i) Tensile or longitudinal or Normal stress :-

If deforming force is applied normally i.e. right angle to the surface area of the body the applied force per unit area is called Tensile or longitudinal or Normal stress.

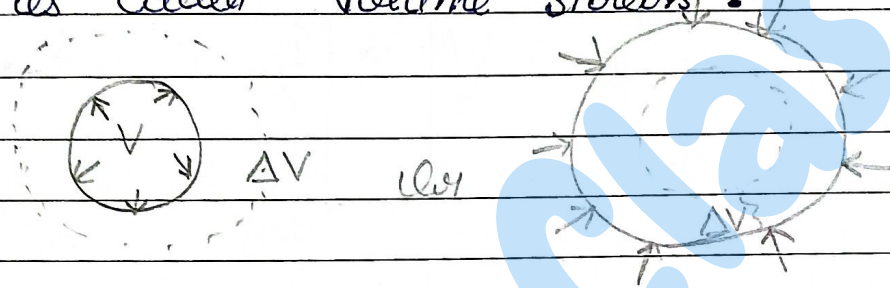
i.e. $\text{Stress} = \frac{F}{A} = \frac{mg}{\pi r^2}$



(ii) Volume or Compressional Stress :-

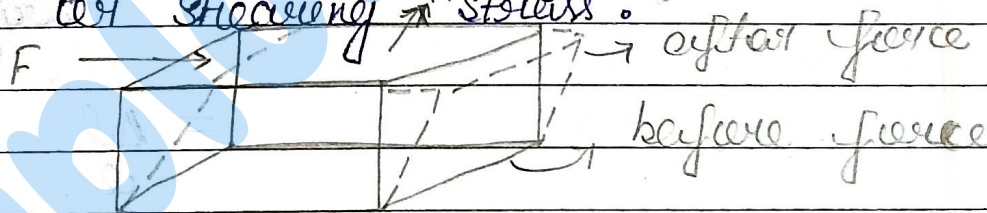
If deforming force is applied normally and it changes volume of the object, is called volume stress. Then force per unit area is called volume stress.

Pressure = $\frac{F}{A}$



(iii) Tangential Stress :-

Restoring force develops tangentially to the surface of body due to change in the shape of the body with change in volume. The stress in the body is called tangential stress or shearing stress.



Tangential stress = $\frac{F}{A}$

* Strain :-

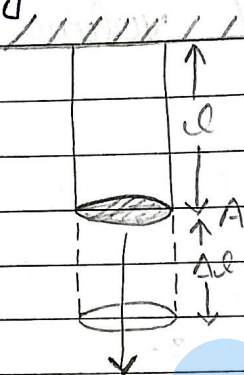
It is the ratio of change in dimension to the original dimension. When deforming force is applied.

Strain = $\frac{\text{Change in dimension}}{\text{Original dimension}}$

It has no unit and no dimension.

* Types of Strain :-

- (i) Longitudinal Strain :- When the deforming force produces change in length alone the strain produced is called longitudinal strain. It is measured by the ratio of change in length of the body to its original length.



If 'l' is the original length & Δl is the change in length.

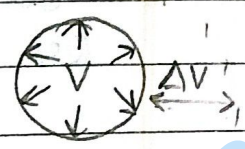
$$\therefore \text{Longitudinal Strain} = \frac{\text{Change in length}}{\text{Original length}}$$

$$= \frac{\Delta l}{l}$$

* Types of Strain :-

(ii) Volumetric Strain :- The ratio of Change in Volume by Original Volume

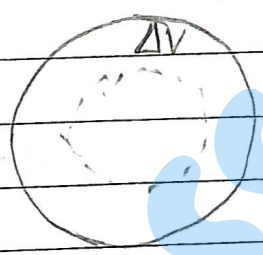
i.e.
$$\text{Volumetric Strain} = \frac{\Delta V}{V}$$



Decrease in Volume :- If Change in Volume is -ve i.e. Volume is decreases then,

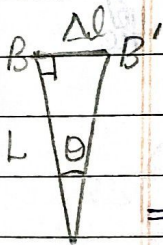
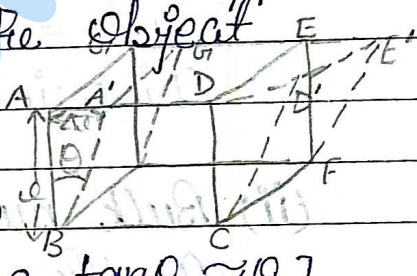
Then,

$$\text{Volumetric Strain} = -\frac{\Delta V}{V}$$



(iii) Shearing Strain :- It is the ratio of Change in shapes to the original shapes. When we applied a force deforming force on the object of there is a change in volume is called shearing strain.

From figure $\text{Shearing Strain} = \tan \theta$
 i.e. $\tan \theta \approx \theta$
 [∵ Where θ is very very small ∴ where]



⇒
$$\text{Shearing Strain} = \frac{BB'}{AB} = \frac{\Delta l}{L}$$

* Hooke's Law :- Within elastic limit stress is directly proportional to strain.

i.e. $\text{Stress} \propto \text{Strain}$	Unit = N/m^2
$\frac{\text{Stress}}{\text{Strain}} = \text{Constant}$	Dimension = $[ML^{-1}T^{-2}]$

This Constant is called ~~modulus~~ modulus of elasticity.

It is of three types :-

- (i) Young's Modulus of elasticity (γ) :- It is defined as the ratio of Normal or tensile or longitudinal stress to the Normal or tensile or longitudinal strain.

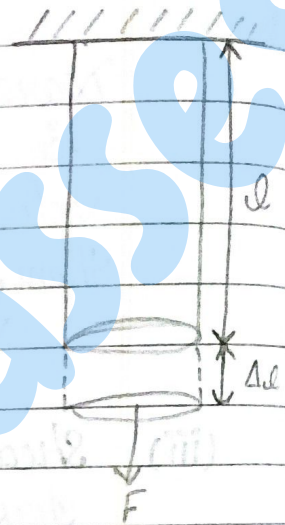
i.e. $\gamma = \frac{\text{Longitudinal stress}}{\text{Longitudinal strain}}$

$$\gamma = \frac{F/A}{\Delta l/l} = \frac{F l}{A \Delta l}$$

$$= \frac{mg l}{\pi r^2 \Delta l}$$

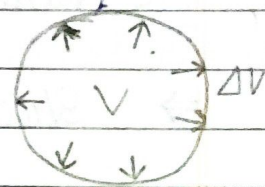
$$\text{Unit} = \text{N/m}^2 = \text{kg m}^{-1} \text{s}^{-2}$$

$$\text{Dimension} = [M^1 L^{-1} T^{-2}]$$



- (ii) Bulk modulus of elasticity (K) :- It is defined as the ratio of Volumetric stress to the Volumetric strain.

i.e. $K = \frac{\text{Volumetric stress}}{\text{Volumetric strain}}$

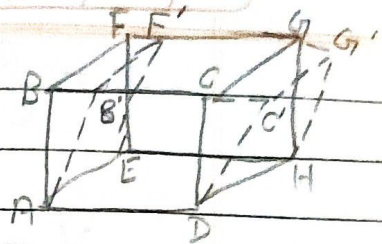


$$K = \frac{F/A}{\Delta V/V} = \frac{F V}{A \Delta V}$$

- (iii) Modulus of rigidity or Rigidity modulus of elasticity (η) :-

It is defined as the ratio of shearing stress to the shearing strain.

i.e. $\eta = \frac{\text{shearing stress}}{\text{shearing strain}}$



$$= \frac{F/A}{\tan \theta} = \frac{F/A}{\frac{BB'}{AB}} = \frac{F/A}{\frac{\Delta l}{l}}$$

$$= \frac{FL}{A\Delta l}$$

Unit = $\text{N/m}^2 = \text{dyne cm}^{-2}$

* Relation Between γ , k and η :-

$$\frac{\gamma}{3k} + \frac{\gamma}{\eta} = 3$$

$$\Rightarrow \frac{1}{3k} + \frac{1}{\eta} = \frac{3}{\gamma}$$

$$\Rightarrow \frac{\eta + 3k}{3k\eta} = \frac{3}{\gamma}$$

$$\Rightarrow \gamma = \frac{9k\eta}{\eta + 3k}$$

* Numericals :-

Q1. A load of 2kg produces an extension of 1mm in a wire 3m in length and 1mm in diameter. Calculate the young's modulus of elasticity of the wire.

Solved - Given,

$$m = 2\text{kg}$$

$$\Delta L = 1\text{mm} = 1 \times 10^{-3} \text{ m}$$

$$L = 3\text{m}$$

$$d = 1\text{mm} = 1 \times 10^{-3} \text{ m}$$

CLASSTIME Pg. No. _____
Date ____/____/____

$$r = \frac{d}{2} = \frac{1 \times 10^{-3}}{2} = 0.5 \times 10^{-3} \text{ m.}$$

$$\gamma = \frac{FL}{A\Delta L} = \frac{mg \times L}{\pi r^2 \Delta L} = \frac{2 \times 9.8 \times 7 \times 100}{11 \times 22 \times 0.5 \times 0.5 \times 10^{-3} \times 10^{-3} \times 10}$$

$$= \frac{98 \times 7 \times 4 \times 10^3 \times 10^3}{11 \times 10}$$

$$= \frac{98 \times 7 \times 4 \times 10^5}{11} \quad \begin{array}{r} 2058(1) \\ 11 \overline{) 2058} \\ \underline{196} \\ 98 \end{array}$$

$$\therefore \text{ m bar } = 248 \times 10^5 \text{ m}$$

$$= \frac{2 \times 9.8 \times 7}{22 \times (0.5 \times 10^{-3})^2} \quad \begin{array}{r} 98 \\ 21 \overline{) 98} \\ \underline{196} \\ 2058 \end{array}$$

$$= \frac{2 \times 9.8 \times 7}{22 \times 1 \times 10^{-6}} \quad \begin{array}{r} 98 \\ 21 \overline{) 98} \\ \underline{196} \\ 2058 \end{array}$$

$$= \frac{2 \times 9.8 \times 3 \times 7}{11 \times (0.5 \times 10^{-3})^2} = \frac{2 \times 9.8 \times 3 \times 7}{11 \times 1 \times 10^{-6} \times 10^{-3}}$$

$$= \frac{9.8 \times 3 \times 7}{11 \times 1 \times 10^{-6} \times 10^{-3}}$$

$$= \frac{9.8 \times 3 \times 7}{11 \times 10^{-9}}$$

$$= \frac{98 \times 3 \times 7 \times 10^{10}}{11}$$

$$= \frac{2 \times 9.8 \times 3 \times 7}{11 \times 22 \times (0.5 \times 10^{-3})^2 \times 1 \times 10^{-3}}$$

$$= \frac{9.8 \times 3 \times 7 \times 10^6}{11 \times 0.25 \times 10^{-6} \times 10^{-3} \times 10}$$

$$= \frac{9.8 \times 3 \times 7 \times 10}{11 \times 25 \times 10^{-9}}$$

$$= \frac{98 \times 3 \times 7 \times 10^{10}}{11 \times 25}$$

Q2/ What force is required to stretch steel wire to double its length when its area of cross section is 1 cm^2 and young's modulus of elasticity is $2 \times 10^{11} \text{ N/m}^2$?

Solution - $A = 1 \text{ cm}^2 \Rightarrow 10^{-4} \text{ m}^2$

$$Y = 2 \times 10^{11}$$

$$L = l$$

$$\Delta L = 2l - l = l$$

$$Y = \frac{FL}{A\Delta L}$$

$$F = \frac{YA\Delta L}{L}$$

$$= \frac{2 \times 10^{11} \times 10^{-4} \times (2l - l)}{l}$$

$$= 2 \times 10^{11} \times 10^{-4} \times (2 - 1) \times 10^0$$

$$= \frac{2 \times 10^{11} \times 10^{-4}}{2 \times 10^7} \text{ N}$$

Q3: 2 wires made of the same material have their lengths in the ratio of 3:4 and their diameters are in the ratio 2:3 if both of them are stretched by the same load what will be the ratio of their extension?

Given,

$$L_1 = 3, L_2 = 4$$

$$D_1 = 2, D_2 = 3$$

$$r_1 = \frac{D_1}{2} = 1, r_2 = \frac{3}{2}$$

$$F_1 = F_2 = F$$

$$\frac{r_1}{r_2} = \frac{1 \times 2}{3} = \frac{2}{3}$$

$$Y = \frac{FL}{A\Delta L}$$

$$\Delta L = \frac{FL}{AY}$$

$$A_1 = \pi r_1^2 = \frac{22}{7} \times (1)^2 = \frac{22}{7}$$

$$A_2 = \pi r_2^2 = \frac{22}{7} \times \frac{3}{2} \times \frac{3}{2} = \frac{99}{14}$$

$$Y_1 = Y_2 = Y$$

$$\Delta L_1$$

$$\frac{F L_1}{A_1 Y_1}$$

$$\Delta L_2$$

$$\frac{F_2 L_2}{A_2 Y_2}$$

$$\frac{F \times 3 \times 7}{22 \times Y}$$

$$\frac{F \times 4 \times 14}{99 \times Y}$$

$$\frac{F L_1}{A_1 Y_1} \times \frac{A_2 Y_2}{F_2 L_2}$$

$$\frac{\Delta L_1}{\Delta L_2} = \frac{F \times 3 \times \frac{99}{14} \times Y \times 7}{\frac{22}{7} \times Y \times \frac{14}{2} \times F \times 4}$$

$$\frac{\Delta L_1}{\Delta L_2} = \frac{27}{16}$$

$$L_1 = 3, L_2 = 4$$

$$D_1 = 2, D_2 = 3$$

$$r_1 = \frac{2}{2} = 1, r_2 = \frac{3}{2}$$

$$F_1 = F_2 = F$$

$$\frac{r_1}{r_2} = \frac{1 \times 2}{3} = \frac{2}{3}$$

$$A_1 = \pi r_1^2 = \frac{22}{7} \times (1)^2 = \frac{22}{7}$$

$$A_2 = \pi r_2^2 = \frac{22}{7} \times \frac{3}{2} \times \frac{3}{2} = \frac{99}{14}$$

Both wires are made up of same material $\gamma_1 = \gamma_2 = \gamma$

Therefore, young's modulus of elasticity for both the wires is same.

$$\gamma_1 = \frac{F_1 L_1}{A_1 \Delta L_1} \quad \text{--- (i)}, \quad \gamma_2 = \frac{F_2 L_2}{A_2 \Delta L_2} \quad \text{--- (ii)}$$

eq. (i) & (ii)

$$\frac{\gamma_1}{\gamma_2} = \frac{F}{F} \quad \Delta L_1 = \frac{F_1 L_1}{A_1 \gamma_1}, \quad \Delta L_2 = \frac{F_2 L_2}{A_2 \gamma_2}$$

From eq. (i) & (ii),

$$\frac{\Delta L_1}{\Delta L_2} = \frac{F_1 L_1}{A_1 \gamma_1}$$

$$= \frac{F_2 L_2}{A_2 \gamma_2}$$

$$= \frac{27}{16}$$

Then ratio will be

$$\Delta L_1 : \Delta L_2 = 27 : 16$$

$$= \frac{F_1 L_1}{A_1 \gamma_1} \times \frac{A_2 \gamma_2}{F_2 L_2}$$

$$\frac{Y_s}{Y_r} = \frac{\Delta L_s}{\Delta L_r}$$

from eq. (3)

$$\therefore \Delta L_r > \Delta L_s$$

$$\therefore Y_s > Y_r$$

Therefore steel is more elastic than rubber.

Q A steel rail is 20m long and has an area of cross section 40cm^2 . Suppose between summer and winter its length changes by 1cm. It is held in winter with force parallel to its length is necessary to keep it from increasing the length in summer ($Y = 9 \times 10^{10} \text{ N/m}^2$).

$$L = 20\text{m}$$

$$\Delta L = 1\text{cm} = 10^{-2}\text{m}$$

$$A = 40\text{cm}^2 = 40 \times 10^{-4}\text{cm}^2$$

$$Y = 9 \times 10^{10}$$

$$F = ?$$

$$\text{Strain} = \frac{\Delta L}{L} = \frac{10^{-2}}{20} = 0.5 \times 10^{-3}$$

$$\text{Stress} = Y \times \text{Strain}$$

$$= 9 \times 10^{10} \times 0.5 \times 10^{-3}$$

$$= 9.5 \times 10^7 \text{ Nm}^{-2}$$

$$\text{Stress} = \frac{F}{A} \Rightarrow F = \text{Stress} \times A$$

$$F = 9.5 \times 10^7 \times 40 \times 10^{-4}$$

$$F = 3.800 \times 10^5 \text{ N}$$

$$F = 3.8 \times 10^5 \text{ N}$$

Date
18/10/24

Q4.6 Which is more elastic rubber or steel?

Ans Steel is more elastic than rubber.

Let us consider two identical wires made up of steel and rubber.

$$A_s = A_r = A$$

$$L_s = L_r = L$$

If we apply some deforming force on both the wire then extension (change in length) in both wire will be ΔL_s & ΔL_r respectively.

$$\text{where } \Delta L_r > \Delta L_s$$

Therefore,

Young's modulus of elasticity for both the wire will be,

$$Y_s = \frac{F}{A_s} \times \frac{L}{\Delta L_s}$$

$$Y_s = \frac{F \times L}{A_s \times \Delta L_s} \quad \text{--- (I)}$$

$$Y_r = \frac{F \times L}{A_r \times \Delta L_r} \quad \text{--- (II)}$$

$$\frac{Y_s}{Y_r} = \frac{\frac{F \times L}{A_s \times \Delta L_s}}{\frac{F \times L}{A_r \times \Delta L_r}} \quad \text{--- (III)}$$

$$\frac{Y_s}{Y_r} = \frac{\frac{F \times L}{A \times \Delta L_s}}{\frac{F \times L}{A \times \Delta L_r}}$$